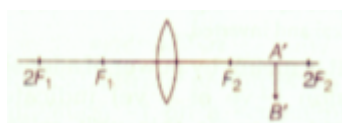


LIGHT -06

Class 10 - Science

1. One-half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? [3]
Verify your answer experimentally. Explain your observations.
2. Observe the following incomplete ray diagram of an object where the image A'B' is formed after refraction from a convex lens. [3]



On the basis of above information fill in the blanks.

- i. The position of object AB would have been...
 - ii. Size of the object would have been ... than the size of image.
- a. Define focal length of a divergent lens. [3]
 - b. A divergent lens of focal length 30 cm forms the image of an object of size 6 cm on the same side as the object at a distance of 15 cm from its optical centre. Use lens formula to determine the distance of the object from the lens and the size of the image formed.
 - c. Draw a ray diagram to show the formation of image in the above situation.
4. Sudha finds out that the sharp image of window pane of her science laboratory is formed at a distance of 15 cm from the lens. She now tries to focus the building visible of her outside the window instead of the window pane without disturbing the lens. In which direction will she move the screen to obtain a sharp image of the building? What is the approximate focal length of this lens? [3]
5. What is atmospheric refraction? Explain with the help of a labelled diagram that the position of a star as seen by us is not its true position. [3]
6. The refractive indices 1.0003, 1.31 1.5 respectively of Air, Ice and Benzine in which of these does the light travels fastest? [3]
7. An object is kept at a distance of 18 cm, 20 cm and 30 cm, from a lens of power +5D. (i) In which case or cases would you get a magnified image? (ii) Which of the magnified image can we get on a screen? (b) List two widely used applications of a convex lens. [3]
- i. Name the spherical mirror used as: [3]
 - a. shaving mirror
 - b. Rear view mirror in vehicles
 - c. Reflection in search-light.
 - ii. Write any three difference between a real and a virtual image.
9. Differentiate between virtual image formed by a concave mirror and of a convex mirror. [3]

10. How can you distinguish between plane mirror, convex mirror and concave mirror by merely looking at the image formed in each case? [3]
11.
 - i. What should be the position of the object when a concave mirror is to be used [3]
 - a. as a shaving mirror and
 - b. in torches producing parallel beam of light?
 - ii. A man standing in front of a mirror, finds his image having a very small head and legs of normal size. What type of mirrors are used in designing such a mirror?
12. A pencil when dipped in water in a glass tumbler appears to be bent at the interface of air and water. Will the pencil to be bent to the same extent, if instead of water we use liquids like, kerosene or turpentine? Support your answer with reasons. [3]
13. Distinguish between real image and virtual image. [3]
14. "A concave mirror of focal length f can form a magnified, erect as well as an inverted image of an object placed in front of it." Justify this statement stating the position of object with respect to the mirror in each case for obtaining these images. [3]
15. How are the power and focal length of a lens related? You are provided with two lenses of focal length 20 cm and 40 cm respectively. Which lens will you use to obtain more convergent light? [3]
16. What should be the position of an object with respect to focus of a convex lens of focal length 20cm, so that its real and magnified image is obtained? [3]
17. Distinguish between a convex lens and a concave lens. [3]
18. A student wants to project the image of a candle flame on a screen 80 cm in front of a mirror by keeping the candle flame at a distance of 20 m from its pole. [3]
 - i. Which type of mirror should the student use?
 - ii. Find the magnification of the image produced.
 - iii. Find the distance between the object and its image.
19. A thin converging lens form a real magnified image and virtual magnified image of an object in front of it. [5]
 - i. Write the positions of the objects in each case.
 - ii. Draw ray diagrams to show the image formation in each case.
 - iii. How will the following be affected on cutting this lens into two halves along the principal axis?
 - a. Focal length
 - b. Intensity of the image formed by half lens.
20. Form the image in case an object is moved from infinity to the concave mirror. [5]
21.
 - i. Define focal length of a spherical lens. [5]
 - ii. A divergent lens has a focal length of 30 cm. At what distance should an object of height 5 cm from the optical centre of the lens be placed, so that its image is formed 15 cm away the lens? Find the size of the image also.
 - iii. Draw a ray diagram to show the formation of image in the above situation.
22.
 - a. List four characteristics of the images formed by plane mirrors. [5]
 - b. A 5 cm tall object is placed at a distance of 20 cm from a concave mirror of focal length 30 cm. Use mirror formula to determine the position and size of the image formed.
23.
 - i. Draw a ray diagram to show the formation of image by a convex lens when an object is placed in front of the lens between its optical centre and principal focus. [5]

- ii. In the above ray diagram mark the object distance (u) and the image distance (v) with their proper signs (+ve or -ve as per the new cartesian sign convention) and state how these distances are related to the focal length (f) of the convex lens in this case.
- iii. Find the power of convex lens which forms a real and inverted image of magnification -1 of an object placed at a distance of 20 cm from its optical centre.
24. Rishi went to a palmist to show his palm. The palmist used a special lens for this purpose. [5]
- State the nature of the lens and the reason for its use.
 - Where should the palmist place/hold the lens so as to have a real and magnified image of an object?
 - If the focal length of this lens is 10 cm and the lens is held at a distance of 5 cm from the palm, use lens formula to find the position and size of the image.
25. a. Name the type of mirror that should be used to obtain the following types of images: [5]
- A magnified and virtual image of an object
 - A diminished and virtual image of an object
- b. Draw labelled ray diagrams to justify your answers in each case mentioned above.
- c. Which of these mirrors can also be used to obtain a magnified and real image of an object? State the position of the object in this case.
26. An object 2 cm high is placed at a distance of 16 cm from a concave mirror which produce a real image 3 cm high. [5]
- What is the focal length of the mirror ?
 - Find the position of the image.
27. An object 2 cm high is placed at a distance of 16 cm from a concave mirror which produces a real image 3 cm high. [5]
- Find the position of the image.
 - What is the focal length of mirror?
28. An object of height 4.0 cm is placed at a distance of 30 cm from optical centre **O** of a concave lens of focal length 20 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre **O** and principal focus **F** on the diagram. Also, find the approximate ratio of size of the image to the size of the object. [5]
29. i. Define optical centre of spherical lens. [5]
- A divergent lens has a focal length of 20 cm. At what distance should an object of height 4 cm from the optical centre of the lens be placed, so that its image is formed 10 cm away from the lens. Find the size of the image also.
 - Draw a ray diagram to show the formation of image in above situation.
30. a. Draw a ray diagram for showing the convergence of a parallel beam of light by a concave mirror and mark the positions of pole and the centre of curvature in it. [5]
- b. An object 4 cm in size is placed at 25 cm in front of a concave mirror of focal length 15 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Find the nature and size of the image.
- c. List any two uses of a concave mirror.
31. Find the size, nature and position of image formed when an object of size 1 is placed at a distance of 15 cm from a concave mirror of focal length 10 cm. [5]
32. Draw a ray diagram in each of the following cases to show the formation of image, when the object is placed: [5]

- i. between the optical center and principal focus of a convex lens.
- ii. anywhere in front of a concave lens.
- iii. at 2F of a convex lens.

State the signs and values of magnifications in the above-mentioned cases (i) and (ii).

33. a. If the image formed by a mirror for all positions of the object placed in front of it, it is always diminished, erect and virtual, state the type of the mirror and also draw a ray diagram to justify your answer. Write one use such mirrors are put to and why? [5]
- b. Define the radius of curvature of spherical mirrors. Find the nature and focal length of a spherical mirror whose radius of curvature is + 24 cm.

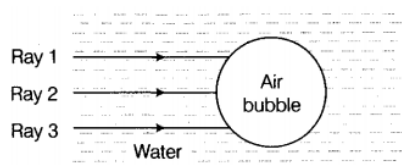
34. What are the rules to form image of an object by concave lens? Form the images of an object when it is moved from infinity to the lens. [5]

35. A student has three concave mirrors A, B and C of focal lengths 20 cm, 15 cm and 10 cm, respectively. For each concave mirror, he performs the experiment of image formation for three values of object distance of 30 cm, 10 cm and 20 cm. [5]

Give reasons for the following:

- i. For the three object distances, identify the mirror which will form an image equal in size to that of object. Find at least one value of object distance.
- ii. Out of the three mirrors, identify the mirror which would be preferred to be used for shaving purpose.
- iii. For the mirror B, draw ray diagram for image formation for any two given values of object distance.

36. An air bubble in water is shown in the figure. Three rays of light are incident on the air bubble. [5]

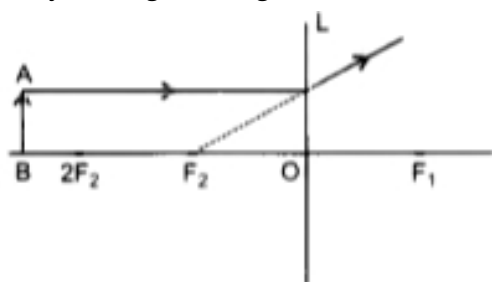


The angle of incidence of ray 1 on the air bubble is greater than the critical angle. The angle of incidence of ray 2 on the air bubble is less than the critical angle. Ray 3 is perpendicular to the surface of the bubble.

- i. In figure at the point where ray 1 meets the air bubble, mark
 - a. the normal to the surface
 - b. the angle of incidence
 - ii. Complete the ray diagram to show how all three rays continue after they meet the air bubble.
 - iii. Define refractive index of water. If the speed of light in air is $3 \times 10^8 \text{ ms}^{-1}$ and the speed of light in water is $2.2 \times 10^8 \text{ ms}^{-1}$. Calculate the refractive index of water.
37. A student wants to project the image of a candle flame on the walls of school laboratory by using a lens. [5]
- i. Which type of lens should he use and why?
 - ii. At what distance in terms of focal length F of the lens should he place the candle flame, so as to get
 - a. a magnified and
 - b. a diminished image respectively, on the wall?
 - iii. Draw ray diagrams to show the formation of the image in each case.
38. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. [5]
- i. Use the lens formula to find the distance of the image from the lens.
 - ii. List four characteristics of the image (nature, position, size, erect/inverted) formed by the lens in this case.

iii. Draw a ray diagram to justify your answer to the part(ii).

39. An object 5 cm high is placed at a distance of 10 cm from a convex mirror of radius of curvature 30 cm. Find the nature, position and size of the image. [5]
40. One-half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? Verify your answers experimentally. Explain your observations. [5]
41. i. A concave mirror of focal length 10 cm can produce a magnified real as well as virtual image of an object placed in front of it. Draw ray diagrams to justify this statement. [5]
 ii. An object is placed perpendicular to the principal axis of a convex mirror of focal length 10 cm. The distance of the object from the pole of the mirror is 10 cm. Find the position of the image formed.
42. An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm. Draw the ray diagram and find the position, size and the nature of the image formed. [5]
43. An object 4.0 cm in size, is placed 25.0 cm in front of a concave mirror of focal length 15.0 cm. [5]
 i. At what distance from the mirror should a screen be placed in order to obtain a sharp image?
 ii. Find the size of the image.
 iii. Draw a ray diagram to show the formation of an image in this case.
44. Study the diagram in figure and answer the following questions: [5]



- a. Name the lens L.
 b. What are the points F_1 and F_2 called?
 c. Complete the diagram to form the image of the object AB.
 d. State three characteristics of the image formed.
45. i. One half of a convex lens of focal length 10 cm is covered with a black paper. Can such a lens produce an image of a complete object placed at a distance of 30 cm from the lens? Draw a ray diagram to justify your answer. [5]
 ii. A 4 cm tall object is placed perpendicular to principal axis of a convex lens of focal length 20 cm. The distance of the object from the lens is 15 cm. Find the nature, position and the size of the image.
46. A student places a candle flame at a distance of about 60 cm from a convex lens of focal length 10 cm and focuses the image of the flame on a screen. After that he gradually moves the flame towards the lens and each time focuses the image on the screen. [5]
 i. In which direction: toward or away from the lens, does he move the screen to focus the image?
 ii. How does the size of the image change?
 iii. How does the intensity of the image change as the flame moves towards the lens?
 iv. Approximately for what distance between the flame and the lens, the image formed on the screen is inverted and of the same size?
47. i. Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab. [5]

- ii. Calculate the refractive index of the material of a glass slab. Given that the speed of light through the glass slab is 2×10^8 m/s and in air is 3×10^8 m/s.
- iii. Calculate the focal length of a lens, if its power is - 2.5 D.
48. It is desired to obtain an erect image of an object, using concave mirror of focal length of 12 cm. [5]
- What should be the range of distance of an object placed in front of the mirror?
 - Will the image be smaller or larger than the object? Draw ray diagram to show the formation of image in this case.
 - Where will the image of this object be, if it is placed 24 cm in front of the mirror? Draw ray diagram for this situation also to justify your answer. Show the positions of pole, principal focus and the centre of curvature in the above ray diagrams.
49. A convex lens of focal length 20 cm can produce a magnified virtual as well as real image. Is this a correct statement? If yes, where shall the object be placed in each case for obtaining these images? [5]
50. We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror? What is the nature of the image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case. [5]
51. Write laws of refraction. Explain the same with the help of ray diagram, when a ray of light passes through a rectangular glass slab. [5]
52. What are rules for formation of images? [5]
53. Define the principal focus of concave mirror. [5]
54. i. Two lenses have power of [5]
- + 2 D
 - 4 D.
- What is the nature and focal length of each lens?
- ii. An object is kept at a distance of 100 cm from each of above lenses. Calculate
- image distance and
 - magnification in each of the two cases.
55. Differentiate between a concave mirror and a convex mirror. [5]
56. i. Define the following terms : [5]
- Power of a lens
 - Principal focus of a concave mirror
- ii. Write the relationship among the object distance (u), image distance (v) and the focal length (f) of a
- Spherical lens
 - Spherical mirror
- iii. An object is placed at a distance of 10 cm from optical centre of a convex lens of focal length 15 cm. Draw a labelled ray diagram to show the formation of image in this case.
57. Under what condition in an arrangement of two plane mirrors, incident ray and reflected ray will always be parallel to each other, whatever may be angle of incidence. Show the same with the help of diagram. [5]
58. Define magnification of a spherical mirror. What will be the magnification in case of plane mirror? [5]
59. A 2.0 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm. The distance of the object from the lens is 15 cm. Find the nature, position, and size of the image. Also, find its magnification. [5]

60. A student focussed the image of a candle flame on a white screen using a convex lens. He noted down the position of the candle screen and the lens as under Position of candle = 12.0 cm Position of convex lens = 50.0 cm Position of the screen = 88.0 cm [5]
- What is the focal length of the convex lens?
 - Where will the image be formed if he shifts the candle towards the lens at a position of 31.0 cm?
 - What will be the nature of the image formed if he further shifts the candle towards the lens?
 - Draw a ray diagram to show the formation of the image in case (iii) as said above.
61. How are the images formed when an object is moved from infinity to the convex lens? [5]
62. An object 1 m tall is placed on the principal axis of a convex lens and its 40 cm tall image is formed on the screen placed at a distance of 70 cm from the object. What is the focal length of the lens? [5]
63. How are the images formed in convex mirror when object is moved from infinity to the mirror? [5]
64. An object 1 cm high is placed on the axis and 15 cm from a concave mirror of focal length 10 cm. Find the position, nature, magnification and size of the image. [5]
65. What are Cartesian sign conventions used in spherical mirror ? What is the mirror formula ? [5]
66. A student wants to project the image of a candle flame on the walls of the school laboratory by using a mirror. [5]
- Which type of mirror should he use and why?
 - At what distance, in terms of focal length f of the mirror, should he place the candle flame to get the magnified image on the wall?
 - Draw a ray diagram to show the formation of the image in this case.
 - Can he use this mirror to project a diminished image of the candle flame on the same wall State 'how' , if your answer is 'yes' and why not', if your answer is 'no'.
67. a. An object is placed in front of a convex lens of focal length f . If the distance of the object from the lens is $2f$, draw a ray diagram to show the formation of the image. State two characteristic of image formed. [5]
- b. A student has focussed the image of a candle flame on a white screen using a convex lens. The situation is as given below:
- Length of the flame = 1.5 cm
- Focal length of the lens = 12 cm
- Distance of the flame from the lens = 18 cm
- If the flame is perpendicular to the principal axis of the lens, calculate the values of the following:
- Distance of the image from the lens
 - Length of the image formed
68. a. What is a lens? List two main categories of lenses. In which category is a double concave lens placed? [5]
- b. A convex lens of focal length 15 cm forms a real image at a distance of 20 cm from its optical centre. Find the position of the object. Is the image formed by the lens magnified or diminished?
69. What is lens formula ? Give its sign conventions and assumptions. [5]
70. i. A person suffering from myopia (near-sightedness) was advised to wear the corrective lens of power -2.5 D. A spherical lens of the same focal length was taken in the laboratory. At what distance should a student place an object from this lens so that it forms an image at a distance of 10 cm from the lens? [5]
- ii. Draw a ray diagram to show the position and nature of the image formed in the above case.
71. i. What is meant by power of a lens? Define its SI unit. [5]

- ii. You have two lenses A and B of focal lengths $+10\text{ cm}$ and -10 cm , respectively. State the nature and power of each lens. Which of the two lenses will form a virtual and magnified image of an object placed 8 cm from the lens? Draw a ray diagram to justify your answer.